method is to treat the Newtonian part of Langevin dynamics by a symplectic method, and follow this by an Ornstein-Uhlenbeck exact solve; this is just what is done in the schemes denoted ⟦BABO⟧ and ⟦ABAO⟧. Thus we could give these two methods the alternate (and more descriptive) names Geometric Langevin (Velocity-Verlet) and Geometric Langevin (Position-Verlet) respectively.

Building on these schemes, the OU solves can instead be placed in the center of the step, or we may choose to have more than one OU solve distributed through the iteration. As it happens, some of these methods are of particular interest. We refer to the method obtained by inserting the OU solve in the middle of the Verlet method either by the strings ⟦ABOBA⟧ (Position-Verlet version) and ⟦BAOAB⟧ (Velocity-Verlet version) or by the more pronounceable names Symmetric Langevin Position-Verlet or Symmetric Langevin Velocity-Verlet, respectively.

For reference and to remove any ambiguity, we write out some of these algorithms below as single timestep maps (sample implementations in MATLAB are available at http://www.MolecularDynamics.info).

⟦ABAO⟧

$$\begin{aligned}
\boldsymbol{q}_{n+1/2} &= \boldsymbol{q}_n + (h/2)\boldsymbol{M}^{-1}\boldsymbol{p}_n,\\
\boldsymbol{p}_{n+1/2} &= \boldsymbol{p}_n - h\nabla U(\boldsymbol{q}_{n+1/2}),\\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_{n+1/2} + (h/2)\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2},\\
\boldsymbol{p}_{n+1} &= e^{-h\gamma}\boldsymbol{p}_{n+1/2} + \zeta_2\boldsymbol{M}^{1/2}\mathbf{R}_n,
\end{aligned}$$

⟦BABO⟧

$$\begin{aligned}
\boldsymbol{p}_{n+1/2} &= \boldsymbol{p}_n - (h/2)\nabla U(\boldsymbol{q}_n),\\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2},\\
\hat{\boldsymbol{p}}_{n+1/2} &= \boldsymbol{p}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_{n+1}),\\
\boldsymbol{p}_{n+1} &= e^{-h\gamma}\hat{\boldsymbol{p}}_{n+1/2} + \zeta_2\boldsymbol{M}^{1/2}\mathbf{R}_n,
\end{aligned}$$

⟦ABOBA⟧

$$\begin{aligned}
\boldsymbol{q}_{n+1/2} &= \boldsymbol{q}_n + (h/2)\boldsymbol{M}^{-1}\boldsymbol{p}_n,\\
\boldsymbol{p}_{n+1/2} &= \boldsymbol{p}_n - (h/2)\nabla U(\boldsymbol{q}_{n+1/2}),\\
\hat{\boldsymbol{p}}_{n+1/2} &= e^{-h\gamma}\boldsymbol{p}_{n+1/2} + \zeta_2\boldsymbol{M}^{1/2}\mathbf{R}_n,\\
\boldsymbol{p}_{n+1} &= \hat{\boldsymbol{p}}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_{n+1/2}),\\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_{n+1/2} + (h/2)\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1},
\end{aligned}$$

⟦BAOAB⟧

$$\begin{aligned}
\boldsymbol{p}_{n+1/2} &= \boldsymbol{p}_n - (h/2)\nabla U(\boldsymbol{q}_n),\\
\boldsymbol{q}_{n+1/2} &= \boldsymbol{q}_n + (h/2)\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2},\\
\hat{\boldsymbol{p}}_{n+1/2} &= e^{-h\gamma}\boldsymbol{p}_{n+1/2} + \zeta_2\boldsymbol{M}^{1/2}\mathbf{R}_n,\\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_{n+1/2} + (h/2)\boldsymbol{M}^{-1}\hat{\boldsymbol{p}}_{n+1/2},\\
\boldsymbol{p}_{n+1} &= \hat{\boldsymbol{p}}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_{n+1}),
\end{aligned}$$

⟦OABAO⟧

$$\begin{aligned}
\boldsymbol{p}_{n+1/2} &= e^{-h\gamma/2}\boldsymbol{p}_n + \zeta_1\boldsymbol{M}^{1/2}\mathbf{R}_n,\\
\boldsymbol{q}_{n+1/2} &= \boldsymbol{q}_n + (h/2)\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2},\\
\hat{\boldsymbol{p}}_{n+1/2} &= \boldsymbol{p}_{n+1/2} - h\nabla U(\boldsymbol{q}_{n+1/2}),\\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_{n+1/2} + (h/2)\boldsymbol{M}^{-1}\hat{\boldsymbol{p}}_{n+1/2},\\
\boldsymbol{p}_{n+1} &= e^{-h\gamma/2}\hat{\boldsymbol{p}}_{n+1/2} + \zeta_1\boldsymbol{M}^{1/2}\mathbf{R}_{n+1/2},
\end{aligned}$$

⟦OBABO⟧

$$\begin{aligned}
\boldsymbol{p}_{n+1/2} &= e^{-h\gamma/2}\boldsymbol{p}_n + \zeta_1\boldsymbol{M}^{1/2}\mathbf{R}_n,\\
\hat{\boldsymbol{p}}_{n+1/2} &= \boldsymbol{p}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_n),\\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\hat{\boldsymbol{p}}_{n+1/2},\\
\hat{\boldsymbol{p}}_{n+1} &= \hat{\boldsymbol{p}}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_{n+1}),\\
\boldsymbol{p}_{n+1} &= e^{-h\gamma/2}\hat{\boldsymbol{p}}_{n+1} + \zeta_1\boldsymbol{M}^{1/2}\mathbf{R}_{n+1/2},
\end{aligned}$$

where $\zeta_j = \left[\mathrm{k_B T}\left(1 - e^{-j\gamma h}\right)\right]^{1/2}$, and $\mathbf{R}_n, \mathbf{R}_{n+1/2}$ are $\mathrm{N_d}$ vectors of standard independent $\mathcal{N}(0,1)$ random variables, resampled at each step. Note that, as in the Velocity-Verlet method, the two apparent evaluations of the force $\boldsymbol{F} = -\nabla U$ in the methods labelled ⟦BABO⟧, ⟦BAOAB⟧ and ⟦OBABO⟧ actually require only a single evaluation at each timestep, since the second evaluation at one step may be reused at the first stage of the following step. Thus all of the above methods require only one force evaluation per timestep, and hence we can think of them having exactly the same computational cost as the evaluation of the force is the principle computational expense.